

Dual Vision Transformer Integration for Race and Gender Recognition Based on Facial Images

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Abstract—Race and gender recognition based on facial images is a part of soft biometrics that has wide applications in identification systems. However, challenges such as dataset bias, variation in expression, pose, and inconsistent race definition remain major obstacles. In this study, the authors propose the integration of two Vision Transformer (ViT) models to improve the accuracy of race and gender classification. The first model, ViT-Face, is trained on the VGGFace2 dataset to extract static facial structure features, while the second model, ViT-Emotion, is trained on the FER-2013 dataset to capture dynamic features of facial expressions. Features from both models are combined and classified using Support Vector Machine (SVM) with parameter optimization through grid search. The dataset used is DemogPairs with 10,800 face images in six balanced classes. The experimental results show that the combination of features from ViT-Face and ViT-Emotion gives the best performance with an accuracy of 0.92, a precision of 0.92, a recall of 0.92, an F1-Score of 0.92, and an ROC AUC of 0.9948. This multi-domain feature integration has proven to be more effective than using a single feature. This approach opens up opportunities for developing smarter, fairer, and more inclusive face-based identification systems. In addition to accuracy, this study also highlights fairness across demographic subgroups and discusses ethical implications of AI-driven demographic classification, ensuring the system is more inclusive and responsible.

Keywords—Vision Transformer, soft biometrics, race recognition, gender recognition, SVM, deep learning

I. INTRODUCTION

The development of artificial intelligence (AI) in the field of computer vision has led to significant advancements, particularly in the classification of race and gender based on facial images. This classification is part of the concept of soft biometrics, which refers to physical characteristics or human behavior that help filter individual identities in identification systems, while addressing challenges related to bias and accuracy [1], [2], [3]. Unlike hard biometrics, such as fingerprints, soft biometrics such as race and gender are not unique but are highly useful in initial forensic analysis, public security, and automated monitoring systems..

AI-based facial recognition systems (FRS) can enhance criminal investigations by providing accurate demographic information, even in poor-quality video conditions. Additionally, AI is utilized in the personalization of digital services and demographic-based recommendation systems

[4]. However, race and gender classification faces several challenges, including the abstract and inconsistent definition of race, variations in facial expression, pose, and lighting, as well as dataset biases that tend to favor Caucasians [5], [6]. To mitigate these issues, various approaches such as adaptive margin-based models and causal graph-based attribute translation have been proposed [7], [8]. Convolutional Neural Networks (CNN) have been used for race classification with high accuracy (97%) on the BUPT dataset [9], but they have limitations in capturing global spatial relationships between facial regions [5], [10]. As a result, additional strategies such as data augmentation, preprocessing, and the adoption of more effective architectures are required to address dataset imbalance and bias.

To address these limitations, Vision Transformer (ViT) has emerged as a popular alternative due to its ability to capture global spatial relationships through self-attention mechanisms [11]. ViT models pretrained on specific domains, such as VGGFace and facial expressions, have demonstrated strong feature representation capabilities [2]. However, most of these models rely on single-domain features, which limits their generalization to heterogeneous data [12].

In this study, the authors propose a new approach in race and gender classification by combining features from two pretrained ViT models on different domains:

1. ViT-Face: pretrained on VGGFace dataset to extract facial structure features.
2. ViT-Emotion: pretrained on facial expression dataset to extract dynamic features.

The extracted features are then combined and classified using Support Vector Machine (SVM). The dataset used is DemogPairs which has 6 balanced classes: Asian_Females, Asian_Males, Black_Females, Black_Males, White_Females, White_Males, with a total of 10,800 images.

In summary, this study has two main contributions:

1. Development of a new approach based on multi-domain feature integration from pretrained ViT models.
2. Comparative evaluation between single and combined features in improving the accuracy of race and gender classification using SVM.

Recent studies have emphasized the importance of bias mitigation in soft biometrics, particularly in transformer-based approaches [13], [14], [15]. Despite advances, existing systems often exhibit racial and gender disparities, underscoring the need for fairness-aware models. Building on this, our study contributes by integrating structural and expressive facial features while explicitly considering subgroup performance to ensure inclusivity and responsible AI practices.

II. RELATED WORKS

Recent advancements in race and gender recognition using Convolutional Neural Networks (CNN) have shown significant improvements in accuracy and efficiency. Various models, such as real-time gender recognition systems achieving accuracy rates of up to 98.59% [16], and others achieving classification accuracy between 99.0% and 100% across several datasets [17], have been developed. Additionally, dual-modality systems have been proposed to enhance facial and gender recognition, improving gender recognition accuracy from 91.80% to 93.22% [18]. Furthermore, the Hierarchical Multi-Task Network (HMT-Net) has been introduced, enabling the simultaneous recognition of gender, race, and facial attractiveness, demonstrating advanced performance through joint training [19]. These advancements underscore the evolving capability of CNNs in handling complex recognition tasks across diverse applications.

Meanwhile, the application of Vision Transformers (ViT) in race and gender recognition has also shown significant promise, with an enhanced ViT model for ethnic minority clothing recognition outperforming traditional convolutional networks, achieving a 1.2% increase in accuracy compared to classic ViT models [20]. The Attribute De-Vision Transformer (AD-vit) has also been developed to improve long-term re-identification by focusing on identity-specific features while reducing the effects of temporal variations in appearance [21]. Efforts to mitigate gender classification bias across gender-race groups using generative views and

structured learning have led to improved accuracy and reduced bias, particularly for underrepresented demographics [22]. Collectively, these developments highlight the transformative potential of ViT in addressing the complex challenges associated with race and gender recognition.

However, despite the effective performance of ViT in gender classification across various datasets such as AFAD and UTKface [23], these models may encode biases related to race and gender, influenced by factors such as training data and model architecture [24]. Techniques such as counterfactual augmentation have shown promise in reducing these biases, although they do not fully eliminate them [24]. Innovative approaches like FaceSaliencyAUG, aimed at enhancing data diversity and reducing gender bias in ViT, have shown improvements in model fairness [25]. While Vision Transformers demonstrate superior feature extraction, most existing research relies on single-domain features and lacks subgroup fairness evaluation, leaving a critical gap addressed in this study. Therefore, this research proposes the integration of multi-domain features and fairness evaluation to address existing biases and improve model outcomes, which remains a significant challenge in this field.

III. METHODS

Figure 1 illustrates the overall flow of the proposed method for race and gender detection based on facial images. The process begins with the collection of face images from the DemogPairs dataset. These images are then subjected to a pre-processing stage to prepare them for further analysis. Following pre-processing, feature extraction is performed using the Vision Transformer (ViT) model, which captures both static and dynamic features from the images. Once the features are extracted, the dataset is split into training and testing subsets for classification. The final output of the system is the classification of each face image into the appropriate race-gender labels. Each step in this flow is detailed sequentially in the diagram, providing a clear overview of the entire process from image collection to classification output.

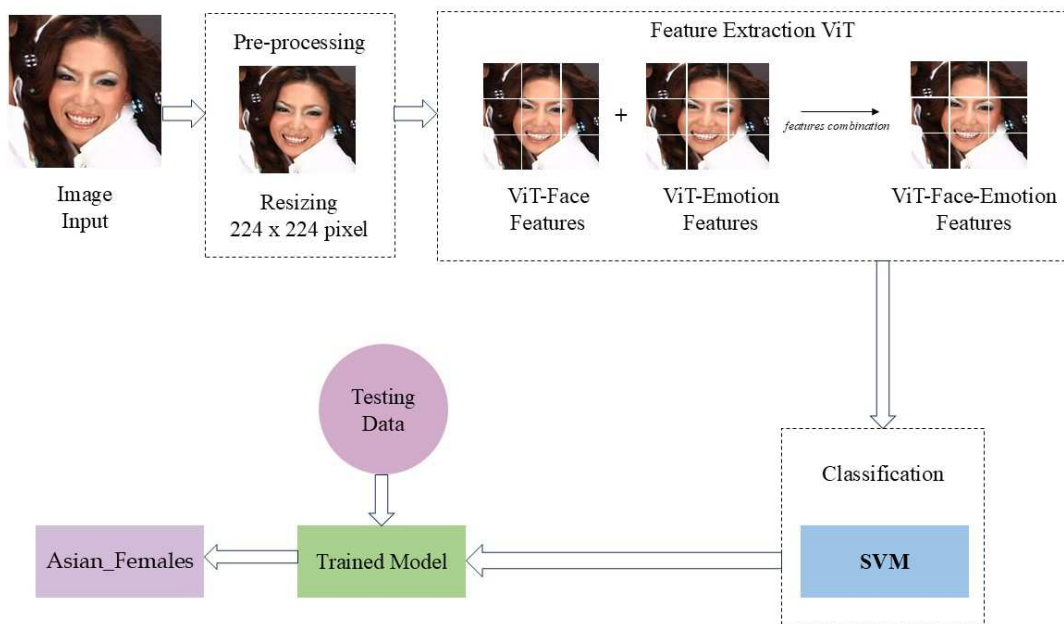


Fig. 1. Proposed Method Diagram

A. Dataset

This study utilizes the DemogPairs dataset, which consists of 10,800 facial images categorized into six balanced demographic groups: Asian-Males, Asian-Females, Black-Males, Black-Females, White-Males, and White-Females [26]. The DemogPairs dataset was introduced by Fernández in the study DemogPairs: Quantifying the Impact of Demographic Imbalance in Deep Face Recognition. From this dataset, approximately 58.3 million facial image pairs were generated for the identity verification task. Each pair is labeled to indicate whether the two images belong to the same individual (genuine) or to different individuals (impostor).

B. Preprocessing

Preprocessing is done to ensure that the image data to be processed gets the same treatment, so that each image in the DemogPairs dataset will be resized to 224x224 pixels. This resizing process is also adjusted to the input size of the Vision Transformer model used in this study.

C. Feature Extraction

This study employs the Vision Transformer (ViT) model for the feature extraction process. ViT is a Transformer-based deep learning architecture introduced by Google Research in a paper by [27]. Unlike traditional convolutional architectures (CNNs), which rely on convolution operations to extract local spatial features, ViT utilizes a self-attention mechanism to capture a global representation of the image. This approach allows ViT to focus on long-range dependencies and global context within the image, offering advantages in certain computer vision tasks.

Figure 2 presents a diagram of the Vision Transformer (ViT) architecture, illustrating how the model processes images for classification tasks. The workflow of the ViT can be explained through several key stages, as follows:

1) *Image Splitting into Patches*: The input image of fixed size ($H \times W \times C$) is divided into several small non-overlapping patches, for example 16×16 pixels. Each patch is then flattened into a one-dimensional vector. If the image is 224×224 , then with a patch size of 16×16 , a total of 196 patches will be produced.

2) *Linear Projection to Patch Embedding*: Each flattened patch vector is projected into a fixed-dimensional space using a linear projection layer. This process aims to transform each patch into a representation vector (embedding) that can be processed by the transformer.

3) *Adding Class Tokens ([CLS] Tokens)*: A special learnable token is added to the embedding sequence and placed at the beginning. This token is known as the [CLS] token and will be used as an aggregate representation of the entire image for classification purposes.

4) *Addition of Positional Embedding*: Since the transformer architecture does not have intrinsic positional information, a positional embedding is added to each patch embedding to preserve the order and spatial position information of each patch in the image.

5) *Transformer Encoder Processing*: A set of embeddings (consisting of patch embedding, [CLS] token, and positional embedding) are then fed into the transformer encoder block. This encoder consists of several layers of

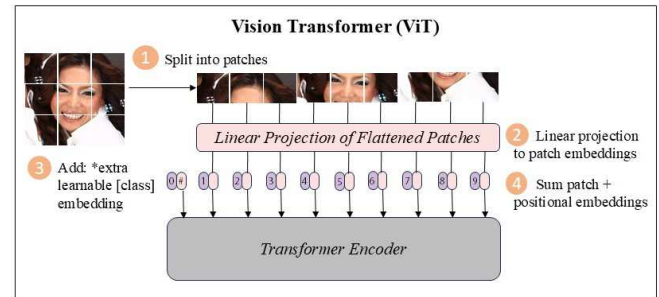


Fig. 2. Vision Transformer (ViT) architecture

multi-head self-attention and feed-forward neural networks. At this stage, the model learns the global feature representation of the image end-to-end.

6) *Output for Classification*: The output of the [CLS] token that has passed through the entire encoder block will be the final representation of the image and is used as input for the classification head (e.g. fully connected layer) to predict the target class, such as identifying race, gender, or facial expression

D. ViT-Face

ViT-Face is a Vision Transformer (ViT) model that has been fine-tuned using the VGGFace2 dataset to enhance performance in face recognition and verification tasks [28]. The model processes 224×224 facial image inputs, which are divided into 16×16 patches. It leverages the self-attention mechanism in the Transformer Encoder to capture spatial relationships globally, offering advantages over traditional Convolutional Neural Networks (CNNs). The output of the model is a 768-dimensional face embedding that can be directly applied to tasks such as face verification, clustering, and retrieval, without the need for further retraining. This is achieved by simply measuring cosine similarity between embeddings.

The VGGFace2 dataset used for pretraining contains 3.31 million images from 9,131 individuals, encompassing a wide range of variations in pose, age, lighting, and ethnicity [29]. This diverse dataset enables ViT-Face to be more robust to variations in visual conditions and environmental noise, enhancing its performance across different scenarios.

E. ViT-Emotion

ViT Facial Emotions Image Detection is a Vision Transformer (ViT) model that has been fine-tuned from the google/vit-base-patch16-224-in21k base architecture specifically for facial emotion classification tasks. By dividing images into small patches and processing them through a self-attention mechanism, this model effectively captures spatial patterns globally, allowing it to detect complex facial expressions more accurately than conventional CNNs.

The model was trained using the FER-2013 dataset, which consists of 35,887 grayscale facial images with a resolution of 48×48 pixels across seven basic emotion categories [30]. Preprocessing steps included parsing the images, converting them to RGB, and applying data augmentation techniques to improve model robustness. Despite the challenges posed by the dataset's low resolution and imbalanced class distribution, ViT Facial Emotions Image Detection demonstrated strong

performance, achieving an accuracy of 91%. This result highlights the model's capability to overcome data limitations and effectively classify complex facial emotions.

F. Classification

The Support Vector Machine (SVM) algorithm is employed for classification tasks, using a linear kernel as its separator function [31], [32]. SVM is a powerful method used for both classification and regression tasks in predictive modeling. The fundamental principle of SVM is to identify the optimal hyperplane that best separates data points into distinct classes. Unlike Neural Networks, which use complex, layered structures to classify data, SVM focuses on finding a clear boundary with the maximum margin between classes.

Initially designed for binary classification ($k = 2$), SVM aims to separate two classes by maximizing the margin, the distance between the hyperplane and the nearest data points from each class. Although SVM was originally limited to linear classification, it has evolved to handle non-linear problems through the use of kernel functions. These kernels enable the transformation of input data into higher-dimensional space, making it possible to solve complex, non-linear classification tasks. Common types of kernel functions used in SVM are listed in Table 1.

IV. RESULTS AND DISCUSSION

A. Experimental Configuration

This section outlines the configuration used for classification tasks, including data settings and training specifications. The study employs the K-Fold Cross Validation technique, where the DemogPairs dataset is divided into multiple subsets. These subsets are alternately used for training and testing, ensuring that each data point is both used for training and testing. The goal of this approach is to evaluate whether the model can effectively train and test on the dataset. In this study, a k value of 5 is used, meaning each proposed ViT model will be trained and tested five times. The final performance is assessed using several quantitative metrics: accuracy, precision, recall, and F1-score.

To analyze the effectiveness of the proposed ViT model integration, a series of experiments will be conducted. The first experiment evaluates the ViT-Face model, followed by a second experiment using the ViT-Emotion model. The final experiment involves the integration of both models, utilizing features from different domains. These experiments will allow for a comprehensive comparison of the model's performance.

B. Performance Comparison

This section presents the results of the experiments conducted based on the configurations described in the previous section. Table 2 summarizes the outcomes of the experiments using three different Vision Transformer (ViT) model configurations: ViT-Face, ViT-Emotion, and a combination of ViT-Face-Emotion.

TABLE I. KERNET TYPE SUPPORT VECTOR MACHINE

Kernel Type	Definition
Linear	$K(x,y) = x.y$
Polynomial	$K(\vec{x_i}, \vec{x_j}) = (\vec{x_i}, \vec{x_j} + 1)^p$
Gaussian	$K(\vec{x_i}, \vec{x_j}) = \exp(-\frac{\ \vec{x_i} - \vec{x_j}\ ^2}{2\sigma^2})$
Sigmoid	$K(\vec{x_i}, \vec{x_j}) = \tanh(\alpha \vec{x_i}, \vec{x_j} + \beta)$

The first experiment utilized ViT-Face, a pretrained model on the VGGFace dataset for extracting facial structure features. This model achieved an accuracy of 90.37%, with balanced precision, recall, and F1-score values of 0.90. The ROC AUC score reached 0.9920, indicating excellent classification performance. The execution time for this model was approximately 8661 seconds, or 2.4 hours.

In the second experiment, ViT-Emotion, pretrained on a facial expression dataset, focused on dynamic features such as changes in the shape of the mouth, eyes, and eyebrows. This model achieved an accuracy of 90.19%, with precision, recall, and F1-score values consistently at 0.90. The ROC AUC score was 0.9916, slightly lower than that of ViT-Face, but the execution time was nearly double, taking 17,653 seconds (4.9 hours).

To further enhance performance, a third experiment combined features from both ViT-Face and ViT-Emotion using feature vector concatenation. This approach resulted in a significant performance improvement, achieving the highest accuracy of 92.41%, with precision, recall, and F1-score values of 0.92, and an ROC AUC score of 0.9948. However, the execution time increased to 20,428 seconds (5.7 hours), indicating the additional computational cost of combining the static and dynamic features.

Overall, the results demonstrate that integrating features from both facial structure and expression significantly improves the performance of the race and gender recognition system. This combined model not only enhances classification accuracy but also ensures greater stability across different race and gender classes.

C. Tuning SVM Parameter

At this stage, the parameter optimization process for the Support Vector Machine (SVM) model is conducted to achieve optimal classification performance. The primary parameters adjusted include the C value, which controls the model's complexity and margin constraints, as well as the kernel type, convergence tolerance (tol), and other relevant parameters such as gamma and degree when using a non-linear kernel. These parameters are fine-tuned through repeated experiments using the grid search method and cross-validation. The specific parameter configurations tested during the tuning process are detailed in Table 3.

TABLE II. ViT MODEL EXPERIMENT RESULTS

Model	Accuracy	Precision	Recall	F1-Score	ROC AUC	Execution Time
ViT-Face	0,90	0,90	0,90	0,90	0,99	8661.936784 seconds
ViT-Emotion	0,90	0,90	0,90	0,90	0,99	17653.006186 seconds
ViT-Face-Emotion	0,92	0,92	0,92	0,92	0,99	20428.116571 seconds

TABLE III. SUPPORT VECTOR MACHINE PARAMETER

Parameter	Values	Description
C Value	[0.1, 1.0, 2.0]	This parameter controls the trade-off between model complexity and classification error on the training data. Higher values impose a greater penalty on misclassification, resulting in a narrower decision margin.
Kernel	['rbf', 'poly']	Two types of non-linear kernels were tested: Radial Basis Function (RBF), which is suitable for data with complex decision boundaries, and polynomial kernel, which captures polynomial relationships among features.
Gamma	['scale', 'auto']	This parameter determines how far the influence of a single training example reaches.
Degree	[2, 3]	Used only for the polynomial kernel; it defines the degree of the polynomial used in the feature transformation.
Tolerance (tol)	[1e-3, 1e-4]	Tolerance for the stopping criterion in the optimization process. Smaller values lead to more precise training but require longer training time.
Random State	[42]	Ensures the reproducibility of experimental results.
Probability	[True]	Enables probability estimates in predictions, which are important for ROC AUC evaluation and probability-based decision-making.
Max Iterations	[1000]	Maximum number of training iterations to prevent overly long optimization processes.

Through cross-parameter combination evaluation, Grid Search identifies the optimal model configuration based on validation metrics, including accuracy, precision, recall, F1-score, and ROC AUC. This process not only improves classification accuracy but also ensures that the model generalizes well to test data, helping to prevent overfitting or underfitting. As shown in Table 4, the results of hyperparameter tuning using Grid Search yield the best configuration for the SVM model with the following

parameters: C = 2.0, kernel = 'poly', degree = 2, gamma = 'scale', tol = 0.001, and max_iter = 1000. This combination enables the model to efficiently capture non-linear relationships while applying a moderate penalty to minimize misclassification. The use of gamma = 'scale' and a moderate convergence tolerance (tol) ensures a stable training process. Overall, this configuration delivers the best classification performance using features extracted by the ViT model, balancing accuracy with the model's generalization ability.

TABLE IV. HYPERPARAMETER TUNING SVM

No	Params	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Accuracy Mean	F1 Score Mean	Precision Mean	Recall Mean	Train Time Mean
1	{'classifier': 'SVC', 'classifier__C': 2.0, 'classifier__degree': 2, 'classifier__gamma': 'scale', 'classifier__kernel': 'poly', 'classifier__max_iter': 1000, 'classifier__probability': True, 'classifier__random_state': 42, 'classifier__tol': 0.0001}	0.93	0.93	0.91	0.92	0.92	0.92	0.92	0.92	0.92	247.7343
2	{'classifier': 'SVC', 'classifier__C': 2.0, 'classifier__degree': 2, 'classifier__gamma': 'scale', 'classifier__kernel': 'poly', 'classifier__max_iter': 1000, 'classifier__probability': True, 'classifier__random_state': 42, 'classifier__tol': 0.001}	0.93	0.93	0.91	0.92	0.92	0.92	0.92	0.92	0.92	238.3589
...	...	...	...	...	...	...	...	...	...	...	...
47	{'classifier': 'SVC', 'classifier__C': 2.0, 'classifier__degree': 3, 'classifier__gamma': 'auto', 'classifier__kernel': 'poly', 'classifier__max_iter': 1000, 'classifier__probability': True, 'classifier__random_state': 42, 'classifier__tol': 0.001}	0.81	0.78	0.78	0.77	0.78	0.78	0.79	0.82	0.78	1331.6959
48	{'classifier': 'SVC', 'classifier__C': 2.0, 'classifier__degree': 3, 'classifier__gamma': 'auto', 'classifier__kernel': 'poly', 'classifier__max_iter': 1000, 'classifier__probability': True, 'classifier__random_state': 42, 'classifier__tol': 0.0001}	0.81	0.78	0.78	0.77	0.78	0.78	0.79	0.82	0.78	1172.5707

TABLE V. CLASSIFICATION ACCURACY BY DEMOGRAPHIC SUBGROUP

Class	Accuracy	Precision	Recall	F1-Score
Black_Males	0.98	0.97	0.94	0.95
White_Females	0.97	0.91	0.91	0.91
Asian_Males	0.97	0.90	0.93	0.92
White_Males	0.98	0.95	0.96	0.95
Black_Females	0.97	0.92	0.91	0.91
Asian_Females	0.97	0.90	0.90	0.90

D. Fairness Evaluation

To complement the overall performance evaluation, we conducted a fairness analysis by examining model accuracy across demographic subgroups. This step is essential to ensure that the proposed approach performs consistently across race and gender categories, rather than favoring only the aggregated results, as shown in Table 5.

Table 5 presents the detailed classification performance for each demographic subgroup, including accuracy, precision, recall, and F1-score. The results indicate that the proposed model achieves stable accuracy values ranging from 0.97 to 0.98 across all subgroups, demonstrating that no particular class is disproportionately advantaged or disadvantaged. Similarly, the precision, recall, and F1-scores remain balanced, with only minor variations between subgroups, suggesting that the model is not biased toward specific race-gender combinations. These findings confirm that the integration of multi-domain features combined with the optimized SVM classifier not only improves the overall performance but also ensures fairness and robustness across demographic categories.

V. CONCLUSION

This study proposes a novel approach to race and gender recognition based on facial images by integrating two Vision Transformer (ViT) models ViT-Face and ViT-Emotion to extract complementary features from static facial structures and dynamic expressions. The extracted features were then combined and classified using a Support Vector Machine (SVM) with optimized hyperparameters. Experimental results on the balanced DemogPairs dataset demonstrated that the integration of multi-domain features significantly improved classification performance, achieving an accuracy of 0.92, precision of 0.92, recall of 0.92, F1-score of 0.92, and an ROC AUC of 0.9948. These findings confirm that combining structural and expressive facial cues enhances the robustness and fairness of demographic classification systems.

Despite these promising results, the study has several limitations. First, the experiments were conducted on a preprocessed and balanced version of the DemogPairs dataset, which may not fully reflect real-world conditions, such as variations in lighting, occlusion, and pose. Second, the computational cost increased significantly when combining features from two different models, potentially limiting its applicability in real-time or resource-constrained environments. Third, the model does not account for non-binary or underrepresented gender identities, which are critical for ensuring inclusivity in modern biometric systems.

For future work, several directions can be explored to enhance the practicality and inclusiveness of the proposed method. One potential improvement is the development of a lightweight, end-to-end integrated ViT model that jointly learns both facial structure and expression features more efficiently. Additionally, incorporating more diverse and inclusive datasets that include non-binary genders and broader ethnic subgroups could help reduce bias and improve generalization. Finally, exploring explainability techniques to identify which facial regions or features contribute most to classification decisions could support transparency and trust in soft biometric systems. This research is conducted solely for academic purposes, with the aim of improving fairness and inclusivity in demographic recognition. The authors affirm that the proposed method should not be used for discriminatory or surveillance practices.

ACKNOWLEDGMENT

The authors thanks to Faculty of Engineering, Universitas Negeri Surabaya, for providing the computational resources and academic support essential to this research. Special thanks are also extended to the research supervisors and colleagues who offered valuable insights and constructive feedback throughout the development of this project.

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